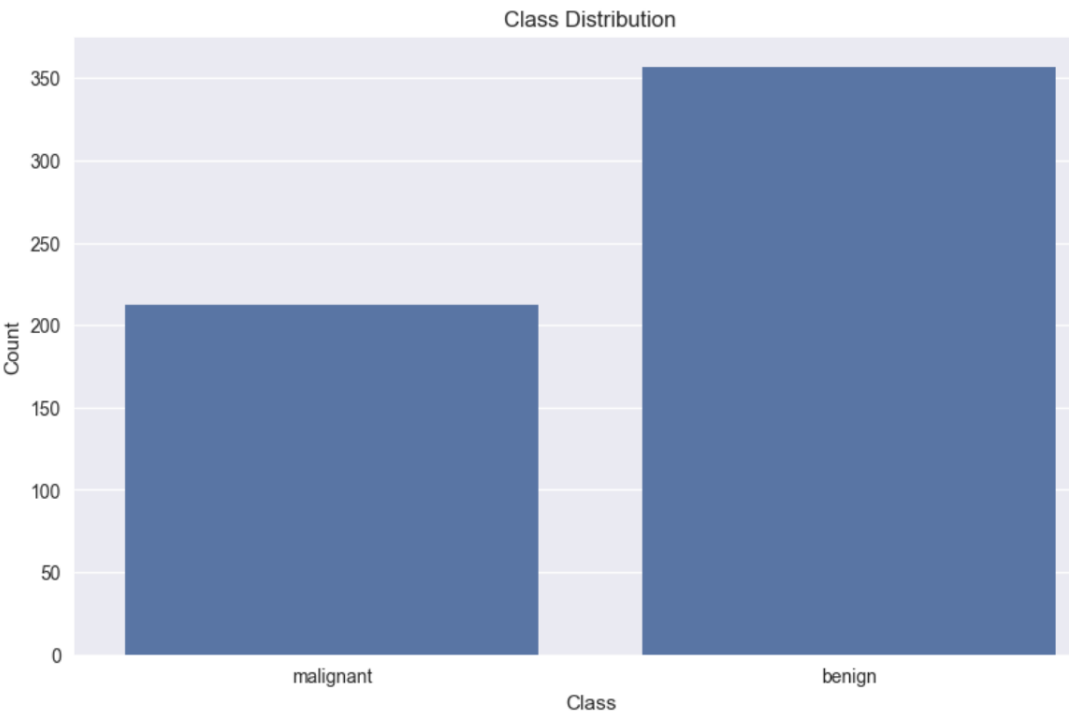
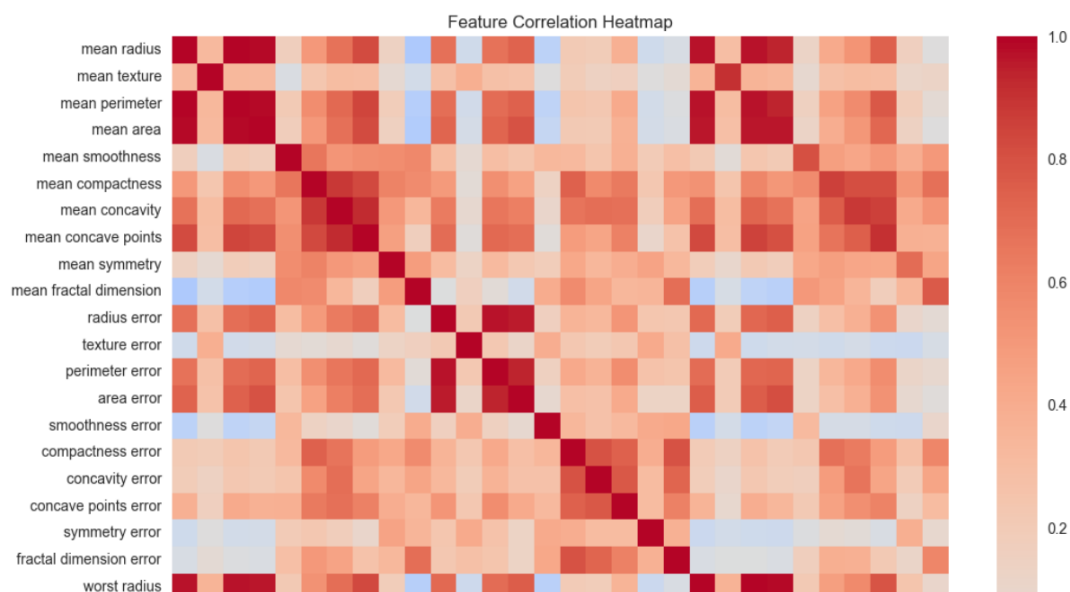


Head :

|   | mean<br>radius | mean<br>texture | mean<br>perimeter | mean<br>area | mean<br>smoothness | mean<br>compactness | mean<br>concavity | mean<br>concave<br>points | mean<br>symmetry | mean<br>fractal<br>dimension | ... | worst<br>radius | worst<br>texture |
|---|----------------|-----------------|-------------------|--------------|--------------------|---------------------|-------------------|---------------------------|------------------|------------------------------|-----|-----------------|------------------|
| 0 | 17.99          | 10.38           | 122.80            | 1001.0       | 0.11840            | 0.27760             | 0.3001            | 0.14710                   | 0.2419           | 0.07871                      | ... | 25.38           | 17.33            |
| 1 | 20.57          | 17.77           | 132.90            | 1326.0       | 0.08474            | 0.07864             | 0.0869            | 0.07017                   | 0.1812           | 0.05667                      | ... | 24.99           | 23.41            |
| 2 | 19.69          | 21.25           | 130.00            | 1203.0       | 0.10960            | 0.15990             | 0.1974            | 0.12790                   | 0.2069           | 0.05999                      | ... | 23.57           | 25.53            |
| 3 | 11.42          | 20.38           | 77.58             | 386.1        | 0.14250            | 0.28390             | 0.2414            | 0.10520                   | 0.2597           | 0.09744                      | ... | 14.91           | 26.50            |
| 4 | 20.29          | 14.34           | 135.10            | 1297.0       | 0.10030            | 0.13280             | 0.1980            | 0.10430                   | 0.1809           | 0.05883                      | ... | 22.54           | 16.67            |

|                        | count | mean       | std        | min        | 25%        | 50%        | 75%        | max        |
|------------------------|-------|------------|------------|------------|------------|------------|------------|------------|
| mean radius            | 569.0 | 14.127292  | 3.524049   | 6.981000   | 11.700000  | 13.370000  | 15.780000  | 28.11000   |
| mean texture           | 569.0 | 19.289649  | 4.301036   | 9.710000   | 16.170000  | 18.840000  | 21.800000  | 39.28000   |
| mean perimeter         | 569.0 | 91.969033  | 24.298981  | 43.790000  | 75.170000  | 86.240000  | 104.100000 | 188.50000  |
| mean area              | 569.0 | 654.889104 | 351.914129 | 143.500000 | 420.300000 | 551.100000 | 782.700000 | 2501.00000 |
| mean smoothness        | 569.0 | 0.096360   | 0.014064   | 0.052630   | 0.086370   | 0.095870   | 0.105300   | 0.16340    |
| mean compactness       | 569.0 | 0.104341   | 0.052813   | 0.019380   | 0.064920   | 0.092630   | 0.130400   | 0.34540    |
| mean concavity         | 569.0 | 0.088799   | 0.079720   | 0.000000   | 0.029560   | 0.061540   | 0.130700   | 0.42680    |
| mean concave points    | 569.0 | 0.048919   | 0.038803   | 0.000000   | 0.020310   | 0.033500   | 0.074000   | 0.20120    |
| mean symmetry          | 569.0 | 0.181162   | 0.027414   | 0.106000   | 0.161900   | 0.179200   | 0.195700   | 0.30400    |
| mean fractal dimension | 569.0 | 0.062798   | 0.007060   | 0.049960   | 0.057700   | 0.061540   | 0.066120   | 0.09744    |
| radius error           | 569.0 | 0.405172   | 0.277313   | 0.111500   | 0.232400   | 0.324200   | 0.478900   | 2.87300    |
| texture error          | 569.0 | 1.216853   | 0.551648   | 0.360200   | 0.833900   | 1.108000   | 1.474000   | 4.88500    |
| perimeter error        | 569.0 | 2.866059   | 2.021855   | 0.757000   | 1.606000   | 2.287000   | 3.357000   | 21.98000   |
| area error             | 569.0 | 40.337079  | 45.491006  | 6.802000   | 17.850000  | 24.530000  | 45.190000  | 542.20000  |
| smoothness error       | 569.0 | 0.007041   | 0.003003   | 0.001713   | 0.005169   | 0.006380   | 0.008146   | 0.03113    |
| compactness error      | 569.0 | 0.025478   | 0.017908   | 0.002252   | 0.013080   | 0.020450   | 0.032450   | 0.13540    |



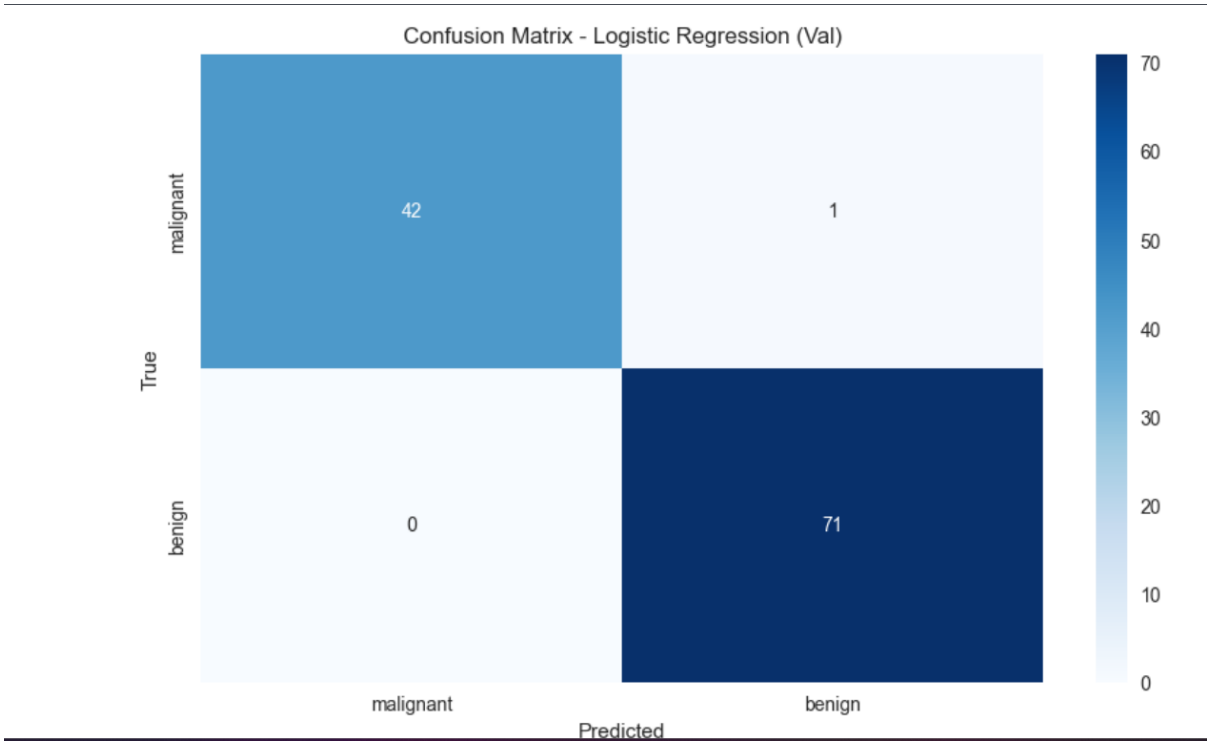


Train/Val/Test sizes: 341/114/114  
 all positive ratio (benign=1): 0.627  
 train positive ratio (benign=1): 0.628  
 val positive ratio (benign=1): 0.623  
 test positive ratio (benign=1): 0.632

Baseline Logistic Regression (Validation):  
 Accuracy: 0.9912280701754386  
 Precision: 0.9861111111111112  
 Recall: 1.0  
 F1: 0.993006993006993

#### Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| malignant    | 1.00      | 0.98   | 0.99     | 43      |
| benign       | 0.99      | 1.00   | 0.99     | 71      |
| accuracy     |           |        | 0.99     | 114     |
| macro avg    | 0.99      | 0.99   | 0.99     | 114     |
| weighted avg | 0.99      | 0.99   | 0.99     | 114     |



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|                                                                                                                                                   |           |        |          |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|----------|---------|
| Validation scores (tuned models):                                                                                                                 |           |        |          |         |
| LR {'acc': 0.9736842105263158, 'precision': 0.9594594594594594, 'recall': 1.0, 'f1': 0.9793103448275862, 'auroc': 0.9967245332459875}             |           |        |          |         |
| SVC {'acc': 0.9736842105263158, 'precision': 0.9594594594594594, 'recall': 1.0, 'f1': 0.9793103448275862, 'auroc': 0.996069439895185}             |           |        |          |         |
| RF {'acc': 0.956140350877193, 'precision': 0.9583333333333334, 'recall': 0.971830985915493, 'f1': 0.965034965034965, 'auroc': 0.9839502129053391} |           |        |          |         |
| Selected best model: LR                                                                                                                           |           |        |          |         |
| Validation Metrics (Best Model):                                                                                                                  |           |        |          |         |
| Accuracy: 0.9736842105263158                                                                                                                      |           |        |          |         |
| Precision: 0.9594594594594594                                                                                                                     |           |        |          |         |
| Recall: 1.0                                                                                                                                       |           |        |          |         |
| F1: 0.9793103448275862                                                                                                                            |           |        |          |         |
| Classification Report:                                                                                                                            |           |        |          |         |
|                                                                                                                                                   | precision | recall | f1-score | support |
| malignant                                                                                                                                         | 1.00      | 0.93   | 0.96     | 43      |
| benign                                                                                                                                            | 0.96      | 1.00   | 0.98     | 71      |
| accuracy                                                                                                                                          |           |        | 0.97     | 114     |
| macro avg                                                                                                                                         | 0.98      | 0.97   | 0.97     | 114     |
| weighted avg                                                                                                                                      | 0.97      | 0.97   | 0.97     | 114     |

Running hyperparameter searches (this is quick but may take ~10-30s)...

Best LR: {'clf\_C': 0.1, 'clf\_penalty': 'l2', 'clf\_solver': 'lbfgs'} CV F1: 0.983800493733981

Best SVC: {'clf\_C': 0.1, 'clf\_gamma': 'scale', 'clf\_kernel': 'linear'} CV F1: 0.9746026613605274

Best RF: {'clf\_n\_estimators': 800, 'clf\_max\_depth': None} CV F1: 0.9791984756284959

Validation scores (tuned models):

LR {'acc': 0.9736842105263158, 'precision': 0.9594594594594594, 'recall': 1.0, 'f1': 0.9793103448275862, 'auroc': 0.967245332459875}

SVC {'acc': 0.9736842105263158, 'precision': 0.9594594594594594, 'recall': 1.0, 'f1': 0.9793103448275862, 'auroc': 0.996069439895185}

RF {'acc': 0.956140350877193, 'precision': 0.9583333333333334, 'recall': 0.971830985915493, 'f1': 0.96503496503496, 'auroc': 0.9839502129053391}

Selected best model: LR

Validation Metrics (Best Model):

Accuracy: 0.9736842105263158

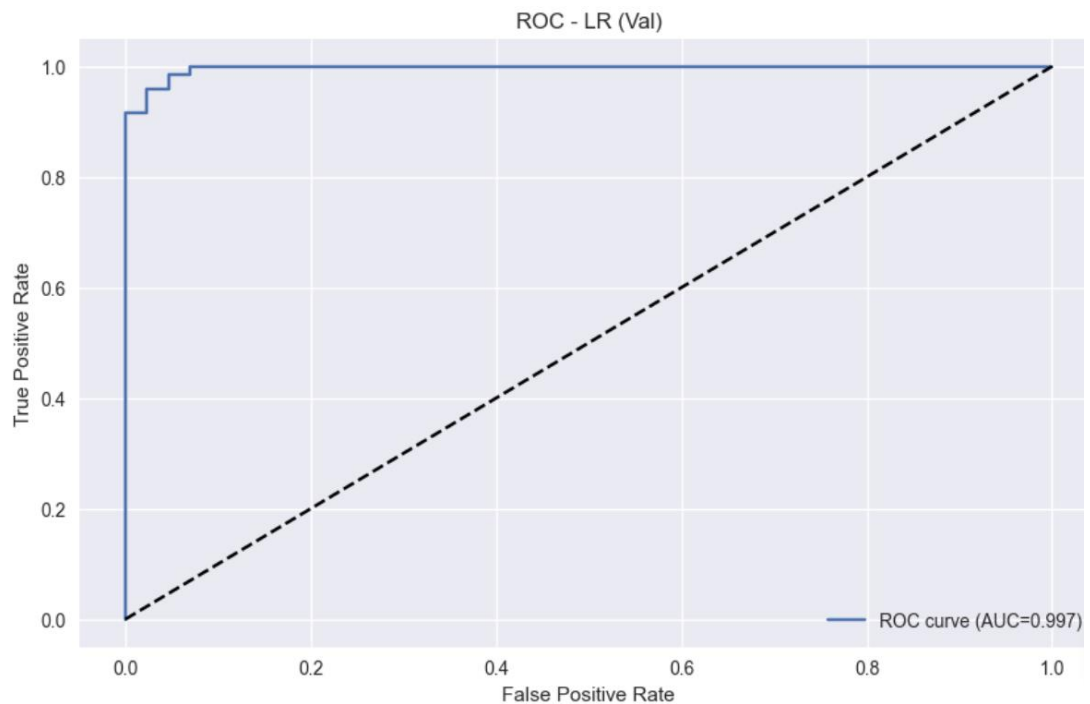
Precision: 0.9594594594594594

Recall: 1.0

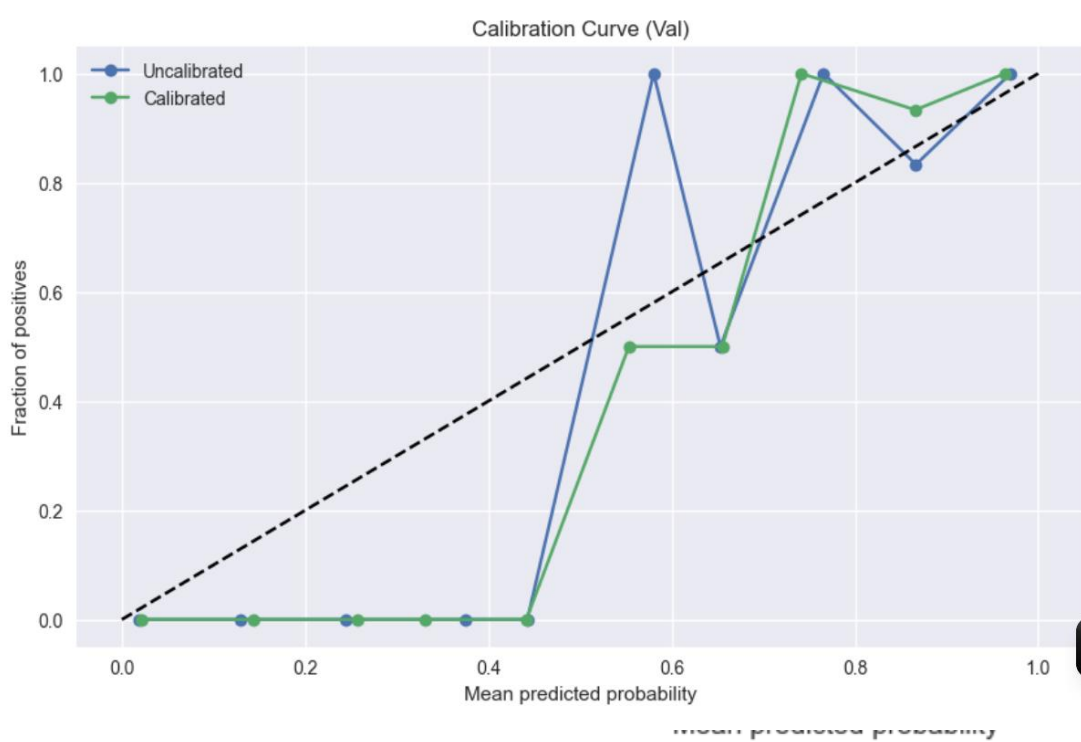
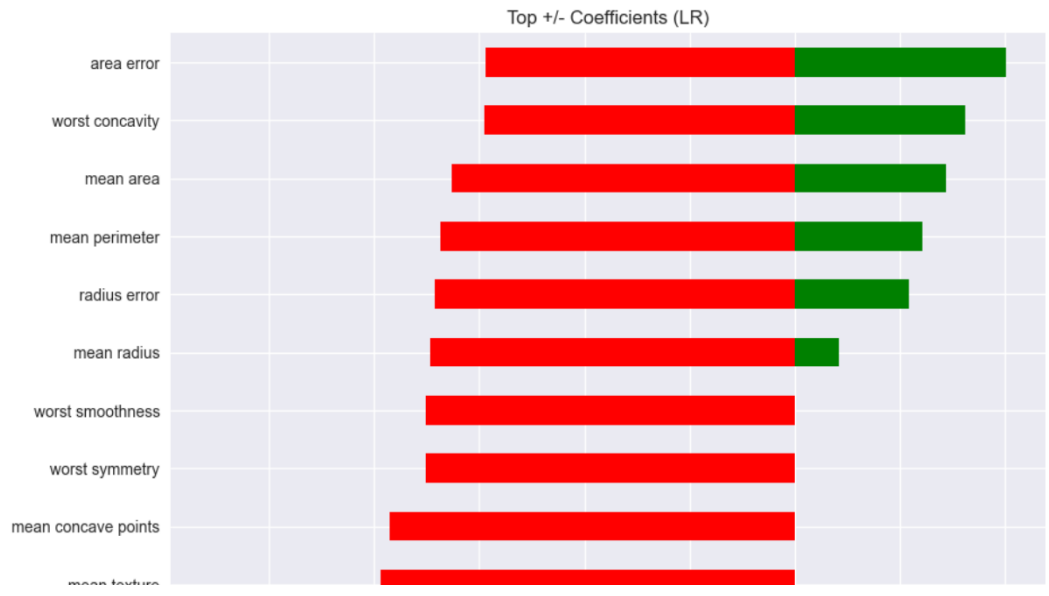
F1: 0.9793103448275862

Classification Report:

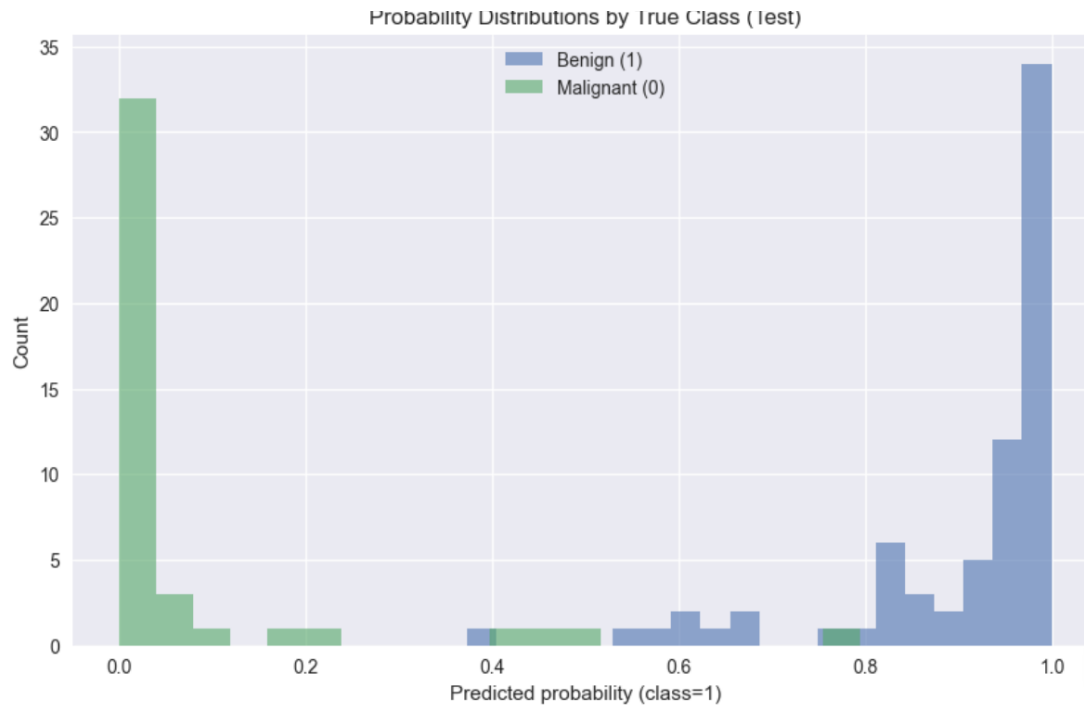
|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| malignant    | 1.00      | 0.93   | 0.96     | 43      |
| benign       | 0.96      | 1.00   | 0.98     | 71      |
| accuracy     |           |        | 0.97     | 114     |
| macro avg    | 0.98      | 0.97   | 0.97     | 114     |
| weighted avg | 0.97      | 0.97   | 0.97     | 114     |



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Test Metrics (Best Model):  
 Accuracy: 0.9736842105263158  
 Precision: 0.9726027397260274  
 Recall: 0.9861111111111112  
 F1: 0.9793103448275862  
 ROC-AUC: 0.9957010582010581  
 Saved model to ../models/breast\_cancer\_clf.joblib  
 Loaded model sanity predictions: [0, 1, 0, 1, 0]  
 SHAP not available; skipping.



Validation: first 15 predictions (True vs Pred with probability)

|    | y_true | y_pred | proba    | correct |
|----|--------|--------|----------|---------|
| 0  | 1      | 1      | 0.945319 | True    |
| 1  | 0      | 0      | 0.000480 | True    |
| 2  | 0      | 0      | 0.018144 | True    |
| 3  | 1      | 1      | 0.990764 | True    |
| 4  | 1      | 1      | 0.925821 | True    |
| 5  | 1      | 1      | 0.998962 | True    |
| 6  | 1      | 1      | 0.669249 | True    |
| 7  | 1      | 1      | 0.904907 | True    |
| 8  | 1      | 1      | 0.995711 | True    |
| 9  | 0      | 0      | 0.000033 | True    |
| 10 | 1      | 1      | 0.994686 | True    |
| 11 | 1      | 1      | 0.953937 | True    |
| 12 | 1      | 1      | 0.995399 | True    |
| 13 | 1      | 1      | 0.982577 | True    |
| 14 | 0      | 0      | 0.013161 | True    |